

PropertyPilot — Unit Test Plan

Project: PropertyPilot (WGU D424 Capstone) **Author:** Michael Riehm **Last updated:** 2026-06-01 **Test framework:** Vitest 4.1.x **Scope:** Backend unit tests (domain, mappers, repositories, controllers, middleware, reports, forecast)

1. Scope and Objectives

1.1 Scope

This test plan covers automated unit tests for the PropertyPilot backend. The suite verifies the building blocks the rest of the system depends on:

- **Domain classes** — validation rules, business invariants, and mutator behavior on `Property` , `Unit` , `Tenant` , `Lease` , `Transaction` , and `MaintenanceTicket` .
- **Mappers** — round-trip conversion between Prisma rows and domain instances.
- **Repositories** — CRUD with owner scoping, paginated list filters, and the search short-circuit.
- **Controllers** — HTTP request handling (success paths, 404 / ownership / validation error paths).
- **Auth middleware** — JWT bearer-token enforcement.
- **Reports** — `RentRoll` , `Profit & Loss` , `Occupancy` , and `Maintenance Aging` row generation.
- **Forecast** — 12-month cash flow projection driven by active leases and trailing expenses.

End-to-end browser flows, deployment pipeline tests, and database integration tests are **out of scope** for this plan — they will be covered manually in the user-guide walkthrough and the deployment screencast.

1.2 Objectives

1. Catch regressions in the core domain rules before they reach the HTTP layer.
2. Confirm that every repository query is scoped to the authenticated owner so that one user can never see or mutate another user's data.
3. Confirm that every controller success path returns the expected payload and that every error path throws the correct error class (so the central error middleware maps it to the right HTTP status).
4. Verify the auth middleware blocks every variant of a missing or malformed `Authorization` header.
5. Confirm that report and forecast aggregations match worked-out expected values from hand-built fixtures.
6. Make the test run reproducible: one command, no external services, no network calls.

1.3 Success criteria

- All test files run green (`npm run test -w backend`) on a clean checkout.
- Every repository CRUD method has at least one success-path test and at least one error-path test.
- Every controller method has at least one success-path test and at least one error-path test.
- The auth middleware has tests for valid token, missing header, malformed header, empty token, and verification failure.

2. Test Approach

2.1 Framework and tooling

- **Vitest 4.1** — fast TypeScript-native test runner, drop-in compatible with Jest assertions but better Vite/TS integration.
- `vi.fn()` **mocks** — used to stand in for Prisma client tables, the auth service, and Express `Request` / `Response` objects.

- **No real database** — every Prisma method is mocked. Tests run in milliseconds and do not require a Postgres container or schema migration.
- `vi.useFakeTimers()` / `vi.setSystemTime()` — used in date-sensitive forecast tests so that the projection window is deterministic regardless of the wall clock.

2.2 Style and conventions

- Tests live next to the source file as `<filename>.test.ts`. The production `tsconfig.build.json` excludes them from the compiled output.
- Each repository test file builds a tiny `tableMock()` helper that returns the subset of Prisma methods the repository actually calls; the whole mocked client is cast via `as unknown as PrismaClient` so the type checker stays happy without us having to mock the full Prisma surface.
- Each controller test file builds a `makeReq()` / `makeRes()` helper. `req.user` is pre-populated with `{ id: 'owner-1', email: 'owner@example.com' }` to simulate the value the auth middleware would attach.
- Domain entity factories use the static `Entity.create({...})` factory for valid cases and the raw constructor for known-invalid cases that need to bypass the factory's auto-validation.

2.3 Coverage strategy

Coverage was driven by the rubric (D.2 specifically calls out repositories, controllers, and auth middleware) plus the OOP requirements from Task 3.B.1 (which puts the domain class hierarchy at the center of the system). The suite therefore stacks four layers of unit tests:

1. Domain — proves the business rules in isolation.
2. Mappers / Repositories — proves the persistence boundary.
3. Controllers — proves the HTTP boundary with the layers below mocked out.
4. Middleware — proves the auth gate that wraps every protected route.

The reports and forecast suites then exercise the higher-level aggregations the dashboard and reporting pages depend on.

2.4 How to run

```
# Run the suite once
npm run test -w backend

# Watch mode (re-runs on change)
npm run test:watch -w backend

# Verbose per-test listing (used for the screenshot in section 4)
npm run test -w backend -- --run --reporter=verbose
```

3. Test Cases

Each table below lists the individual cases in one test file. IDs are stable within a file (`TC-<file-key>-NN`). All cases share the same expected high-level outcome — the assertion passes — so the "Expected output" column captures the specific behavior being asserted.

3.1 Domain layer — `backend/src/domain/`

`Property.test.ts`

ID	Description	Input	Expected output
TC-PROP-01	Validation passes for a fully populated property	Valid Property instance	validate() returns { ok: true }
TC-PROP-02	Validation passes when addressLine2 is null	Property with addressLine2: null	validate() returns { ok: true }
TC-PROP-03	Fails when name is blank	name: ''	validate().ok === false
TC-PROP-04	Fails when name exceeds 120 characters	name over 120 chars	validate().ok === false
TC-PROP-05	Fails when addressLine1 is blank	addressLine1: ''	validate().ok === false
TC-PROP-06	Fails when city is blank	city: ''	validate().ok === false
TC-PROP-07	Fails when state is lowercase	state: 'wi'	validate().ok === false
TC-PROP-08	Fails when state is too long	state: 'WIS'	validate().ok === false
TC-PROP-09	Fails when postal code is malformed	postalCode: 'abc'	validate().ok === false
TC-PROP-10	Accepts ZIP+4 format	postalCode: '53703-1234'	validate() returns { ok: true }
TC-PROP-11	Aggregates multiple errors at once	Multiple invalid fields	errors array contains every offender
TC-PROP-12	rename() updates name and bumps updatedAt	rename('New Name')	name updated, timestamp advances
TC-PROP-13	fullAddress() combines parts with optional line 2	Property with line 2	Joined address string

Unit.test.ts

ID	Description	Input	Expected output
TC-UNIT-01	Validation passes for a typical unit	Valid unit	{ ok: true }
TC-UNIT-02	Passes with null squareFeet	squareFeet: null	{ ok: true }
TC-UNIT-03	Fails when label is blank	label: ''	{ ok: false }

TC-UNIT-04	Fails when bedrooms is negative	bedrooms: -1	{ ok: false }
TC-UNIT-05	Fails when bedrooms is fractional	bedrooms: 2.5	{ ok: false }
TC-UNIT-06	Fails when bathrooms is negative	bathrooms: -1	{ ok: false }
TC-UNIT-07	Fails when marketRent is negative	marketRent: -100	{ ok: false }
TC-UNIT-08	Fails when squareFeet is negative	squareFeet: -10	{ ok: false }
TC-UNIT-09	Fails when propertyId is empty	propertyId: ''	{ ok: false }
TC-UNIT-10	setMarketRent updates value and bumps updatedAt	setMarketRent(1500)	new rent + timestamp advance

Tenant.test.ts

ID	Description	Input	Expected output
TC-TEN-01	Validation passes for a typical tenant	Valid tenant	{ ok: true }
TC-TEN-02	Passes with null phone	phone: null	{ ok: true }
TC-TEN-03	Fails when firstName is blank	firstName: ''	{ ok: false }
TC-TEN-04	Fails when lastName is blank	lastName: ''	{ ok: false }
TC-TEN-05	Fails when email is invalid	email: 'not-an-email'	{ ok: false }
TC-TEN-06	Fails when phone has too few digits	phone: '123'	{ ok: false }
TC-TEN-07	Accepts phone with formatting characters	phone: '(608) 555-1234'	{ ok: true }
TC-TEN-08	fullName() joins first and last with a space	Avery + Lee	'Avery Lee'

Lease.test.ts

ID	Description	Input	Expected output
TC-LEASE-01	Validation passes for a typical lease	12-month active lease	{ ok: true }
TC-LEASE-02	Fails when start equals end	Same date both fields	{ ok: false }
TC-LEASE-03	Fails when start is after end	Reversed dates	{ ok: false }
TC-LEASE-04	Fails when monthlyRent is zero	monthlyRent: 0	{ ok: false }

TC-LEASE-05	Fails when securityDeposit is negative	securityDeposit: -1	{ ok: false }
TC-LEASE-06	Fails when unitId or tenantId is blank	Either id ''	{ ok: false }
TC-LEASE-07	isActiveOn true within window when ACTIVE	Date inside range	true
TC-LEASE-08	isActiveOn false outside window	Date outside range	false
TC-LEASE-09	isActiveOn false when status is PENDING	Status 'PENDING'	false
TC-LEASE-10	termInMonths approximates length	1-year lease	~12
TC-LEASE-11	terminate() flips status and bumps updatedAt	terminate()	status TERMINATED + new timestamp

Transaction.test.ts

ID	Description	Input	Expected output
TC-TX-01	Validation passes for a typical expense	Valid EXPENSE	{ ok: true }
TC-TX-02	Fails when amount is zero	amount: 0	{ ok: false }
TC-TX-03	Fails when description is blank	description: ''	{ ok: false }
TC-TX-04	Fails when propertyId is blank	propertyId: ''	{ ok: false }
TC-TX-05	Fails when RENT_INCOME has no leaseId	Rent w/o lease	{ ok: false }
TC-TX-06	Passes when RENT_INCOME has a leaseId	Rent + lease	{ ok: true }
TC-TX-07	isIncome true for income types	RENT_INCOME / DEPOSIT_INCOME / OTHER_INCOME	true
TC-TX-08	isIncome false for expense and refund	EXPENSE / REFUND	false
TC-TX-09	signedAmount positive for income, negative otherwise	Mixed types	Correct sign

MaintenanceTicket.test.ts

ID	Description	Input	Expected output
TC-MT-01	Validation passes for an open ticket	Open ticket	{ ok: true }
TC-MT-02	Passes for a closed ticket with resolvedAt	Closed + resolvedAt	{ ok: true }
TC-MT-03	Fails when closed without resolvedAt	CLOSED, resolvedAt: null	{ ok: false }
TC-MT-04	Fails when title is blank	title: ''	{ ok: false }
TC-MT-05	Fails when description is blank	description: ' '	{ ok: false }
TC-MT-06	Fails when resolvedAt is before reportedAt	Out-of-order dates	{ ok: false }
TC-MT-07	isOpen true for OPEN / IN_PROGRESS, false otherwise	All four statuses	Correct booleans
TC-MT-08	ageInDays measures from reportedAt to as-of	10-day window	10
TC-MT-09	close() sets status and resolvedAt	close(date)	status CLOSED, resolvedAt set

3.2 Mappers — backend/src/domain/mappers.test.ts

ID	Description	Input	Expected output
TC-MAP-01	Property round-trip preserves every field	Domain → Prisma input → back	Equal field-by-field
TC-MAP-02	Property mapper preserves null addressLine2	line 2 null	line 2 stays null
TC-MAP-03	Unit round-trips numeric fields cleanly	Decimal rent, sqft	No rounding drift
TC-MAP-04	Unit preserves null squareFeet	sqft null	stays null
TC-MAP-05	Tenant round-trips name, email, phone	All three fields	Equal
TC-MAP-06	Lease round-trips dates and currency	start/end + amounts	Equal
TC-MAP-07	Rent transaction round-trips with lease/unit links	Income tx	Equal

TC-MAP-08	Expense round-trips without unit/lease	Expense tx	Equal w/ nulls preserved
TC-MAP-09	Maintenance ticket round-trips with null resolvedAt	Open ticket	Equal
TC-MAP-10	Maintenance ticket round-trips closed with resolvedAt	Closed ticket	Equal

3.3 Repository layer — backend/src/repositories/

PropertyRepository.test.ts

ID	Description	Input	Expected output
TC-RP-PROP-01	findById returns null when no row matches owner scope	id + ownerId, no match	null, where clause includes ownerId
TC-RP-PROP-02	findById returns Property instance when row exists	id + ownerId, match	Property instance
TC-RP-PROP-03	list returns paginated result with defaults	ownerId only	page 1, pageSize 20, total 1
TC-RP-PROP-04	list honors custom page and pageSize	page: 3, pageSize: 10	take: 10, skip: 20
TC-RP-PROP-05	create persists entity and returns mapped instance	Valid Property	Prisma create called, instance returned
TC-RP-PROP-06	create throws DomainValidationError ON invalid entity	Invalid Property	Throws, Prisma not called
TC-RP-PROP-07	update throws NotFoundError when no rows updated	count 0	Throws NotFoundError
TC-RP-PROP-08	update re-reads and returns updated row	count 1	Updated entity returned
TC-RP-PROP-09	delete throws NotFoundError when no rows match	count 0	Throws NotFoundError
TC-RP-PROP-10	delete succeeds when a row is deleted	count 1	Resolves with undefined
TC-RP-PROP-11	search runs case-insensitive OR across name, addressLine1, city	'maple'	3-clause OR, ownerId scoped
TC-RP-PROP-12	search returns empty for blank query without hitting Prisma	' '	[], Prisma not called

UnitRepository.test.ts

ID	Description	Input	Expected output
----	-------------	-------	-----------------

TC-RP-UNIT-01	<code>findById</code> scopes through <code>property.ownerId</code>	<code>id + ownerId</code>	Join clause includes <code>property: { ownerId }</code>
TC-RP-UNIT-02	<code>findById</code> returns null when nothing matches	No row	null
TC-RP-UNIT-03	<code>list</code> joins through property for owner scoping and filters by <code>propertyId</code>	<code>ownerId + propertyId</code>	Both filters applied
TC-RP-UNIT-04	<code>create</code> persists the entity	Valid Unit	Prisma create called
TC-RP-UNIT-05	<code>update</code> throws <code>NotFoundError</code> when count is zero	count 0	Throws
TC-RP-UNIT-06	<code>update</code> re-reads and returns when count is one	count 1	Updated unit
TC-RP-UNIT-07	<code>delete</code> throws <code>NotFoundError</code> when count is zero	count 0	Throws

TenantRepository.test.ts

ID	Description	Input	Expected output
TC-RP-TEN-01	<code>findById</code> scopes directly by <code>ownerId</code>	<code>id + ownerId</code>	Where includes <code>ownerId</code>
TC-RP-TEN-02	<code>findById</code> returns null when missing	No row	null
TC-RP-TEN-03	<code>list</code> returns <code>PaginatedResult</code> with totals + <code>pageSize</code>	<code>ownerId</code>	Correct paged response
TC-RP-TEN-04	<code>update</code> throws <code>NotFoundError</code> when no rows match	count 0	Throws
TC-RP-TEN-05	<code>delete</code> succeeds when count is one	count 1	Resolves
TC-RP-TEN-06	<code>search</code> runs OR across name and email fields	<code>'avery'</code>	OR with 3 fields
TC-RP-TEN-07	<code>search</code> short-circuits on blank query	<code>''</code>	<code>[]</code> , no Prisma call

LeaseRepository.test.ts

ID	Description	Input	Expected output
TC-RP-LEASE-01	<code>findById</code> scopes through <code>unit.property.ownerId</code>	<code>id + ownerId</code>	Two-level join in where clause
TC-RP-LEASE-02	<code>list</code> passes through <code>unitId</code> , <code>tenantId</code> , status filters	All three	All applied

TC-RP-LEASE-03	update throws <code>NotFoundError</code> when no rows match owner	count 0	Throws
TC-RP-LEASE-04	update re-reads and returns when count is one	count 1	Updated lease
TC-RP-LEASE-05	delete throws <code>NotFoundError</code> when no rows match	count 0	Throws

TransactionRepository.test.ts

ID	Description	Input	Expected output
TC-RP-TX-01	<code>findById</code> scopes through <code>property.ownerId</code>	id + <code>ownerId</code>	Join clause includes <code>ownerId</code>
TC-RP-TX-02	<code>findById</code> returns null when not found	No row	null
TC-RP-TX-03	<code>list</code> applies <code>propertyId</code> , <code>unitId</code> , <code>leaseId</code> , type, and date range	All filters	All applied in where
TC-RP-TX-04	<code>list</code> omits date filter when no range provided	No dates	<code>where.date</code> undefined
TC-RP-TX-05	update throws <code>NotFoundError</code> when no rows match	count 0	Throws
TC-RP-TX-06	update re-reads and returns when count is one	count 1	Updated transaction
TC-RP-TX-07	delete throws <code>NotFoundError</code> when no rows match	count 0	Throws
TC-RP-TX-08	search returns empty for blank query without hitting Prisma	<code>' '</code>	<code>[]</code> , no Prisma call
TC-RP-TX-09	search runs case-insensitive <code>contains</code> on description	<code>'rent'</code>	<code>contains</code> clause, <code>ownerId</code> scoped

MaintenanceTicketRepository.test.ts

ID	Description	Input	Expected output
TC-RP-MT-01	<code>findById</code> scopes through <code>property.ownerId</code>	id + <code>ownerId</code>	Join clause includes <code>ownerId</code>
TC-RP-MT-02	<code>list</code> applies <code>propertyId</code> and status filters	Both	Both applied
TC-RP-MT-03	<code>list</code> with <code>openOnly</code> filters to OPEN or IN_PROGRESS	<code>openOnly: true</code>	<code>status: { in: [...] }</code>
TC-RP-MT-04	update throws <code>NotFoundError</code> when no rows match	count 0	Throws

TC-RP-MT-05	update re-reads and returns when count is one	count 1	Updated ticket
TC-RP-MT-06	delete throws NotFoundError when no rows match	count 0	Throws

UserRepository.test.ts

ID	Description	Input	Expected output
TC-RP-USER-01	findById returns user record when found	id, row exists	UserRecord
TC-RP-USER-02	findById returns null when not found	No row	null
TC-RP-USER-03	findByEmail looks up by email field	email	Prisma called with where: { email }
TC-RP-USER-04	findByEmail returns null when not found	No row	null
TC-RP-USER-05	create passes email and passwordHash and returns the record	New user	Prisma create called, record returned

3.4 Controllers — backend/src/controllers/

propertyController.test.ts

ID	Description	Input	Expected output
TC-CTL-PROP-01	list returns paginated payload with serialized rows	Default query	data array w/ fullAddress
TC-CTL-PROP-02	get returns serialized property when found	Existing id	JSON payload
TC-CTL-PROP-03	get throws NotFoundError when missing	Missing id	Throws NotFoundError
TC-CTL-PROP-04	create persists and returns 201	Valid body	res.status(201), payload
TC-CTL-PROP-05	create throws on Zod validation failure	name: ''	Throws
TC-CTL-PROP-06	update throws NotFoundError when missing	Missing id	Throws
TC-CTL-PROP-07	update renames and returns updated payload	{ name: 'Renamed' }	Updated payload
TC-CTL-PROP-08	remove returns 204 on success	Existing id	res.status(204)

unitController.test.ts

ID	Description	Input	Expected output
TC-CTL-UNIT-01	get returns unit when found	Existing id	JSON payload
TC-CTL-UNIT-02	get throws <code>NotFoundError</code> when missing	Missing id	Throws
TC-CTL-UNIT-03	create verifies parent property before creating	Valid body, owner owns property	201 + property lookup
TC-CTL-UNIT-04	create throws <code>NotFoundError</code> when parent property is not the user's	Foreign <code>propertyId</code>	Throws, create not called
TC-CTL-UNIT-05	create rejects invalid body with Zod error	<code>label: ''</code>	Throws
TC-CTL-UNIT-06	update throws <code>NotFoundError</code> when unit missing	Missing id	Throws
TC-CTL-UNIT-07	update updates market rent and returns payload	<code>{ marketRent: 1500 }</code>	Updated payload
TC-CTL-UNIT-08	remove returns 204 on success	Existing id	204

tenantController.test.ts

ID	Description	Input	Expected output
TC-CTL-TEN-01	get returns serialized tenant when found	Existing id	<code>fullName</code> included
TC-CTL-TEN-02	get throws <code>NotFoundError</code> when missing	Missing id	Throws
TC-CTL-TEN-03	create returns 201 on success	Valid body	201 + payload
TC-CTL-TEN-04	create rejects invalid email	<code>email: 'not-email'</code>	Throws
TC-CTL-TEN-05	update throws <code>NotFoundError</code> when missing	Missing id	Throws
TC-CTL-TEN-06	update updates name and returns payload	<code>{ firstName: 'Bee' }</code>	Updated payload
TC-CTL-TEN-07	remove returns 204	Existing id	204

leaseController.test.ts

ID	Description	Input	Expected output
TC-CTL-LEASE-01	get returns lease when found	Existing id	Payload
TC-CTL-LEASE-02	get throws <code>NotFoundError</code> when missing	Missing id	Throws

TC-CTL-LEASE-03	create verifies unit and tenant before creating	Valid body, owner owns both	201 + 2 lookups
TC-CTL-LEASE-04	create throws <code>NotFoundError</code> when unit not owned	Foreign unitId	Throws
TC-CTL-LEASE-05	create throws <code>NotFoundError</code> when tenant not owned	Foreign tenantId	Throws
TC-CTL-LEASE-06	update throws <code>NotFoundError</code> when lease missing	Missing id	Throws
TC-CTL-LEASE-07	update terminates lease and returns payload	{ status: 'TERMINATED' }	Updated payload
TC-CTL-LEASE-08	remove returns 204	Existing id	204

transactionController.test.ts

ID	Description	Input	Expected output
TC-CTL-TX-01	get returns serialized transaction when found	Existing id	Payload with <code>isIncome</code> + <code>signedAmount</code>
TC-CTL-TX-02	get throws <code>NotFoundError</code> when missing	Missing id	Throws
TC-CTL-TX-03	create verifies property, unit, and lease before creating	Valid body	All 3 lookups + 201
TC-CTL-TX-04	create throws <code>NotFoundError</code> when property not owned	Foreign propertyId	Throws
TC-CTL-TX-05	create throws <code>NotFoundError</code> when attached unit not owned	Foreign unitId	Throws
TC-CTL-TX-06	update throws <code>NotFoundError</code> when missing	Missing id	Throws
TC-CTL-TX-07	update updates amount and returns payload	{ amount: 2000 }	Updated payload
TC-CTL-TX-08	remove returns 204	Existing id	204

authController.test.ts

ID	Description	Input	Expected output
TC-CTL-AUTH-01	register creates a user, signs a token, returns 201	Mixed-case email	201, lowercased email, token, no password hash
TC-CTL-AUTH-02	register maps Prisma P2002 to <code>ConflictError</code>	Duplicate email	Throws <code>ConflictError</code>

TC-CTL-AUTH-03	register rejects invalid body	Bad email/short pw	Throws
TC-CTL-AUTH-04	login returns token + user on valid credentials	Valid email + password	200 + token
TC-CTL-AUTH-05	login throws 401 when user not found	Unknown email	Throws <code>HttpError(401)</code>
TC-CTL-AUTH-06	login throws 401 when password mismatch	Wrong password	Throws <code>HttpError(401)</code>
TC-CTL-AUTH-07	me returns user without password hash	Valid req.user	Payload, no passwordHash
TC-CTL-AUTH-08	me throws 401 when authenticated user no longer exists	User deleted	Throws <code>HttpError(401)</code>

dashboardController.test.ts

ID	Description	Input	Expected output
TC-CTL-DASH-01	summary aggregates property/unit counts, occupancy, YTD totals, ticket statuses	Fixture w/ 1 property, 2 units, 1 active lease, mixed txs and tickets	Correct totals + status counts

searchController.test.ts

ID	Description	Input	Expected output
TC-CTL-SEARCH-01	search returns combined hits across properties, tenants, transactions	q: 'maple'	3 hits, correct counts, property names joined onto tx hits
TC-CTL-SEARCH-02	search rejects empty query with Zod error	q: ''	Throws

reportsController.test.ts

ID	Description	Input	Expected output
TC-CTL-REP-01	rentRoll returns report JSON with 'Rent Roll' title	Default	title === 'Rent Roll'
TC-CTL-REP-02	pn1 defaults dateFrom/dateTo to current YTD	Default	title === 'Year-to-date Profit & Loss'
TC-CTL-REP-03	occupancy returns report JSON with 'Occupancy' title	Default	title === 'Occupancy'
TC-CTL-REP-04	maintenanceAging returns report JSON with 'Maintenance Aging' title	Default	title === 'Maintenance Aging'
TC-CTL-REP-05	maintenanceAging rejects invalid asOf date	asOf: 'not-a-date'	Throws Zod error

forecastController.test.ts

ID	Description	Input	Expected output
TC-CTL-FC-01	forecast returns result for owned property	Existing propertyId	12 projections, defaults applied
TC-CTL-FC-02	forecast honors monthsAhead and trailingMonths overrides	monthsAhead=6, trailingMonths=3	6 projections, fields echoed
TC-CTL-FC-03	forecast throws NotFoundError when property not owned	Foreign propertyId	Throws
TC-CTL-FC-04	forecast rejects monthsAhead out of range	monthsAhead=999	Throws Zod error

3.5 Middleware — backend/src/middleware/auth.test.ts

ID	Description	Input	Expected output
TC-MW-AUTH-01	Attaches req.user and calls next() on valid bearer token	Bearer valid.jwt	req.user set, next() called
TC-MW-AUTH-02	Accepts case-insensitive Bearer prefix	bearer abc...	Works the same
TC-MW-AUTH-03	Rejects with 401 when Authorization is missing	No header	next(HttpError(401))
TC-MW-AUTH-04	Rejects with 401 when scheme is not Bearer	Basic abc123	next(HttpError(401))
TC-MW-AUTH-05	Rejects with 401 when bearer token is empty	Bearer	next(HttpError(401))
TC-MW-AUTH-06	Rejects with 401 when token verification throws	Bearer bad	next(HttpError(401))
TC-MW-AUTH-07	Does not leak verification error details to caller	Verifier throws containing secret	err.message does not include leaked text

3.6 Reports — backend/src/reports/**RentRollReport.test.ts**

ID	Description	Input	Expected output
TC-REP-RR-01	Produces one row per lease, joined with property/unit/tenant	Fixture w/ 2 leases	2 rows, joined fields
TC-REP-RR-02	Filters to a single property when propertyId is set	propertyId filter	Only matching rows
TC-REP-RR-03	Respects asOf by excluding leases not covering that date	asOf outside one lease	Excluded

TC-REP-RR-04	Sets title, generatedAt, and toJSON shape	Generated report	Title, timestamp, columns, rows
--------------	---	------------------	---------------------------------

PnLReport.test.ts

ID	Description	Input	Expected output
TC-REP-PNL-01	Groups income and expenses by month and adds a Total row	Mixed-month txs	One row per month + Total
TC-REP-PNL-02	Returns empty rows array when there is no data	Empty txs	rows: []
TC-REP-PNL-03	Reports negative net when expenses exceed income	Expense-heavy month	Net negative
TC-REP-PNL-04	Sets YTD title	Generated	'Year-to-date Profit & Loss'

OccupancyReport.test.ts

ID	Description	Input	Expected output
TC-REP-OCC-01	Counts occupied and vacant units per property and appends Total	Mixed occupancy	Per-property rows + Total
TC-REP-OCC-02	Filters to a single property when propertyId is given	propertyId filter	Only that property
TC-REP-OCC-03	Shows 0% when no leases overlap the asOf date	All leases ended	0% rows

MaintenanceAgingReport.test.ts

ID	Description	Input	Expected output
TC-REP-MA-01	Places each ticket in the correct aging bucket	Tickets of varying ages	Buckets correctly
TC-REP-MA-02	Returns no rows when there are no open tickets	All closed	Empty rows
TC-REP-MA-03	Sorts within the same age bucket by priority	Mixed priorities	High → Low order

3.7 Forecast — backend/src/forecast/CashFlowForecaster.test.ts

ID	Description	Input	Expected output
TC-FC-01	Throws <code>NotFoundError</code> when property is not owned	Foreign propertyId	Throws
TC-FC-02	Produces 12 monthly projections by default	Default options	12 entries

TC-FC-03	Keeps income constant when a lease covers every month	12-month lease	Income flat
TC-FC-04	Drops income to zero once the lease ends	Lease ends mid-window	Zero income after end
TC-FC-05	Starts income at zero when lease begins mid-forecast	Lease starts later	Zero income before start
TC-FC-06	Computes baseline expense as trailing average	Mixed expense history	Correct mean
TC-FC-07	Flags months where expenses exceed income	Sparse income	expensesExceedIncome: true
TC-FC-08	Ignores leases on units that do not belong to the target property	Mixed-property lease set	Only target-property leases included

4. Test Run Evidence

The screenshots below are from the run captured on the day the suite was finalized. They are reproduced in `docs/test-results.md` alongside the pass/fail accounting.

4.1 Run summary

```
npm run test -w backend
```



```
PS C:\Code\Software Engineering Capstone D424\d424-software-engineering-capstone> npm run test -w backend -- --run
```

```
> @propertyilot/backend@0.1.0 test
> vitest run --run
```

```
RUN v4.1.6 C:/Code/Software Engineering Capstone D424/d424-software-engineering-capstone/backend
```

```
> src/controllers/authController.test.ts [queued]
> src/controllers/dashboardController.test.ts [queued]
> src/controllers/forecastController.test.ts [queued]
> src/controllers/leaseController.test.ts [queued]
> src/controllers/propertyController.test.ts [queued]
> src/controllers/reportsController.test.ts [queued]
> src/controllers/searchController.test.ts [queued]
> src/controllers/tenantController.test.ts [queued]
> src/controllers/transactionController.test.ts [queued]
> src/controllers/unitController.test.ts [queued]
> src/domain/Lease.test.ts [queued]
> src/domain/MaintenanceTicket.test.ts [queued]
> src/domain/mappers.test.ts [queued]
> src/domain/Property.test.ts [queued]
> src/domain/Tenant.test.ts [queued]
> src/domain/Transaction.test.ts [queued]
> src/domain/Unit.test.ts [queued]
✓ src/repositories/UserRepository.test.ts (5 tests) 4ms

> src/controllers/authController.test.ts [queued]
> src/controllers/dashboardController.test.ts [queued]
> src/controllers/forecastController.test.ts [queued]
> src/controllers/leaseController.test.ts [queued]
> src/controllers/propertyController.test.ts [queued]
> src/controllers/reportsController.test.ts [queued]
> src/controllers/searchController.test.ts [queued]
> src/controllers/tenantController.test.ts [queued]
> src/controllers/transactionController.test.ts [queued]
> src/controllers/unitController.test.ts [queued]
> src/domain/Lease.test.ts [queued]
> src/domain/MaintenanceTicket.test.ts 0/9
> src/domain/mappers.test.ts [queued]
> src/domain/Property.test.ts [queued]
> src/domain/Tenant.test.ts [queued]
> src/domain/Transaction.test.ts [queued]
✓ src/domain/Transaction.test.ts (9 tests) 5ms
✓ src/domain/MaintenanceTicket.test.ts (9 tests) 7ms
✓ src/reports/PnLReport.test.ts (4 tests) 7ms
✓ src/middleware/auth.test.ts (7 tests) 5ms
✓ src/reports/MaintenanceAgingReport.test.ts (3 tests) 4ms
✓ src/reports/OccupancyReport.test.ts (3 tests) 13ms
✓ src/forecast/CashFlowForecaster.test.ts (8 tests) 10ms
✓ src/domain/Lease.test.ts (11 tests) 26ms
✓ src/domain/Unit.test.ts (10 tests) 26ms
✓ src/domain/Property.test.ts (13 tests) 25ms
✓ src/domain/Tenant.test.ts (8 tests) 5ms
✓ src/domain/mappers.test.ts (10 tests) 6ms
✓ src/reports/RentRollReport.test.ts (4 tests) 13ms
✓ src/controllers/dashboardController.test.ts (1 test) 4ms
✓ src/repositories/LeaseRepository.test.ts (5 tests) 5ms
✓ src/repositories/MaintenanceTicketRepository.test.ts (6 tests) 6ms
✓ src/repositories/TransactionRepository.test.ts (9 tests) 7ms
✓ src/repositories/UnitRepository.test.ts (7 tests) 6ms
✓ src/repositories/TenantRepository.test.ts (7 tests) 8ms
✓ src/repositories/PropertyRepository.test.ts (12 tests) 8ms
```

```
✓ src/controllers/reportsController.test.ts (5 tests) 6ms
✓ src/controllers/tenantController.test.ts (7 tests) 8ms
✓ src/controllers/transactionController.test.ts (8 tests) 9ms
✓ src/controllers/authController.test.ts (8 tests) 9ms
✓ src/controllers/unitController.test.ts (8 tests) 10ms
✓ src/controllers/leaseController.test.ts (8 tests) 11ms
✓ src/controllers/propertyController.test.ts (8 tests) 12ms
✓ src/controllers/forecastController.test.ts (4 tests) 6ms
✓ src/controllers/searchController.test.ts (2 tests) 8ms

Test Files 30 passed (30)
  Tests 209 passed (209)
Start at 10:28:13
Duration 2.38s (transform 8.12s, setup 0ms, import 20.02s, tests 281ms, environment 3ms)

❖ PS C:\Code\Software Engineering Capstone D424\d424-software-engineering-capstone> 
```

4.2 Verbose per-test listing

```
npm run test -w backend -- --run --reporter=verbose
```

```

PS C:\Code\Software Engineering Capstone D424\d424-software-engineering-capstone> $OutputEncoding = [System.Text.Encoding]::UTF8
>> [Console]::OutputEncoding = [System.Text.Encoding]::UTF8
>> chcp 65001 | Out-Null
>> npm run test -w backend -- --run --reporter=verbose 2>&1 | Select-Object -First 60

> @propertyilot/backend@0.1.0 test
> vitest run --run --reporter=verbose

RUN v4.1.6 C:\Code\Software Engineering Capstone D424\d424-software-engineering-capstone/backend

✓ src/domain/Property.test.ts > Property.validate > passes for a fully populated property 2ms
✓ src/domain/Property.test.ts > Property.validate > passes when addressLine2 is null 0ms
✓ src/domain/Property.test.ts > Property.validate > fails when name is blank 1ms
✓ src/domain/Property.test.ts > Property.validate > fails when name exceeds 120 characters 0ms
✓ src/domain/Property.test.ts > Property.validate > fails when addressLine1 is blank 0ms
✓ src/domain/Property.test.ts > Property.validate > fails when city is blank 0ms
✓ src/domain/Property.test.ts > Property.validate > fails when state is lowercase 0ms
✓ src/domain/Property.test.ts > Property.validate > fails when state is too long 0ms
✓ src/domain/Property.test.ts > Property.validate > fails when postalCode is malformed 0ms
✓ src/domain/Property.test.ts > Property.validate > accepts ZIP+4 format 0ms
✓ src/domain/Property.test.ts > Property.validate > aggregates multiple errors at once 0ms
✓ src/domain/Property.test.ts > Property mutators > rename() updates name and bumps updatedAt 14ms
✓ src/domain/Property.test.ts > Property mutators > fullAddress() combines the parts with optional line 2 0ms
✓ src/domain/Unit.test.ts > Unit.validate > passes for a typical unit 3ms
✓ src/domain/Unit.test.ts > Unit.validate > passes with null squareFeet 0ms
✓ src/domain/Unit.test.ts > Unit.validate > fails when label is blank 0ms
✓ src/domain/Unit.test.ts > Unit.validate > fails when bedrooms is negative 0ms
✓ src/domain/Unit.test.ts > Unit.validate > fails when bedrooms is fractional 0ms
✓ src/domain/Unit.test.ts > Unit.validate > fails when bathrooms is negative 0ms
✓ src/domain/Unit.test.ts > Unit.validate > fails when marketRent is negative 0ms
✓ src/domain/Unit.test.ts > Unit.validate > fails when squareFeet is negative 0ms
✓ src/domain/Unit.test.ts > Unit.validate > fails when propertyId is empty 0ms
✓ src/domain/Unit.test.ts > Unit mutators > setMarketRent updates value and bumps updatedAt 14ms
✓ src/domain/Lease.test.ts > Lease.validate > passes for a typical lease 3ms
✓ src/domain/Lease.test.ts > Lease.validate > fails when startDate equals endDate 0ms
✓ src/domain/Lease.test.ts > Lease.validate > fails when startDate is after endDate 0ms
✓ src/domain/Lease.test.ts > Lease.validate > fails when monthlyRent is zero 0ms
✓ src/domain/Lease.test.ts > Lease.validate > fails when securityDeposit is negative 0ms
✓ src/domain/Lease.test.ts > Lease.validate > fails when unitId or tenantId is blank 0ms
✓ src/domain/Lease.test.ts > Lease behavior > isActiveOn returns true within the lease window when status is ACTIVE 0ms
✓ src/domain/Lease.test.ts > Lease behavior > isActiveOn returns false outside the lease window 0ms
✓ src/domain/Lease.test.ts > Lease behavior > isActiveOn returns false when status is PENDING 0ms
✓ src/domain/Lease.test.ts > Lease behavior > termInMonths approximates the length 0ms
✓ src/domain/Lease.test.ts > Lease behavior > terminate() flips status and bumps updatedAt 14ms
✓ src/reports/OccupancyReport.test.ts > OccupancyReport.generate > counts occupied and vacant units per property and appends a Total row 8ms
✓ src/reports/OccupancyReport.test.ts > OccupancyReport.generate > filters to a single property when propertyId is given 0ms
✓ src/reports/OccupancyReport.test.ts > OccupancyReport.generate > shows 0% occupancy when no leases overlap the asOf date 0ms
✓ src/domain/Tenant.test.ts > Tenant.validate > passes for a typical tenant 3ms
✓ src/domain/Tenant.test.ts > Tenant.validate > passes with null phone 0ms
✓ src/domain/Tenant.test.ts > Tenant.validate > fails when firstName is blank 0ms
✓ src/domain/Tenant.test.ts > Tenant.validate > fails when lastName is blank 0ms
✓ src/domain/Tenant.test.ts > Tenant.validate > fails when email is invalid 0ms
✓ src/domain/Tenant.test.ts > Tenant.validate > fails when phone has too few digits 0ms
✓ src/domain/Tenant.test.ts > Tenant.validate > accepts phone with formatting characters 0ms
✓ src/domain/Tenant.test.ts > Tenant.fullName > joins first and last with a space 0ms
✓ src/reports/RentRollReport.test.ts > RentRollReport.generate > produces one row per lease with property, unit, and tenant joined in 9ms
✓ src/reports/RentRollReport.test.ts > RentRollReport.generate > filters to a single property when propertyId is set 0ms
✓ src/reports/RentRollReport.test.ts > RentRollReport.generate > respects an asOf date by excluding leases not covering that date 0ms
✓ src/reports/RentRollReport.test.ts > RentRollReport.generate > sets title, generatedAt, and toJSON shape 0ms
✓ src/domain/Transaction.test.ts > Transaction.validate > passes for a typical expense 2ms
✓ src/domain/Transaction.test.ts > Transaction.validate > fails when amount is zero 0ms
✓ src/domain/Transaction.test.ts > Transaction.validate > fails when description is blank 0ms
✓ src/domain/Transaction.test.ts > Transaction.validate > fails when propertyId is blank 0ms
PS C:\Code\Software Engineering Capstone D424\d424-software-engineering-capstone>

```

5. Maintenance

- New domain or repository code must ship with a matching `*.test.ts` next to the source file.
- Controller tests are required for any new HTTP route — at minimum one success path and one error path.
- The CI command is `npm run test -w backend`; it must finish green before any rubric-tagged commit.